

# Tossaporn (Tree) Saengja

Experienced in developing advanced deep learning models (GenAI, LLMs) and leading research over six years.

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## education

### Massachusetts Institute of Technology

Master of Engineering in Computer Science and Engineering 4.8/5.0 2019-2020  
Bachelor of Science in Computer Science and Engineering 4.8/5.0 2015-2019  
**International Olympiad in Informatics 2013 (IOI)**, Bronze, 85th among 299 contestants (77 countries)  
**ACM Boston Preliminary 2016 (BOSP)**, 4<sup>th</sup> place among 37 teams (16 schools)  
*IOI Thailand Coach (2023-Present)*  
*Teaching Assistant for MIT 6.824: Distributed Systems (Prof. Robert Morris, Spring 2020)*

## experience

**SiData+, Siriraj Hospital** — Data Scientist 2024-Present  
- Developing a novel medical reasoning Large Language Model (LLM) with SCB 10x, focusing on improving diagnostic accuracy for general medical applications.

**PreceptorAI, CARIVA** — AI Consultant 2023-2025  
- Developed machine learning models in collaboration with Siriraj Radiology Department:  
- Multi-label segmentation model (nnUNet, 0.98 Dice) for vertebrae L3 (muscle, fats) quantification in MRI.  
- Vision-Large Language Model to generate diagnostic reports from chest X-ray images (PyTorch).  
- Generative text-to-image model (diffusion model) for synthetic chest X-ray generation (PyTorch, MONAI).  
- Led research and development of Gindee, a VLM based nutrition estimator.

**Vision and Learning Lab, VISTEC** — Research Assistant, under Prof. Supasorn Suwajanakorn 2023-2024  
- Investigated object collocation by developing Score Distillation Sampling (SDS) to leverage Stable Diffusion's priors.  
- Enhanced local attention masking in CLIP for zero-guidance segmentation.  
- Conducted experiments on diffusion model memorization (CIFAR-10).

**Mahidol Wittayanusorn High School** (national top-tier science school) — Computer Science Teacher 2020-2023  
- Coached students to represent Thailand in IOI 2022.  
- Lectured data science (Python), web dev, blockchain, Computer Olympiad (C++), programming (Python).  
- Advised year-long student projects, primarily in machine learning.

**Agnos** — Technical Cofounder 2019-2023  
- Co-developed an AI-powered diagnosis system (Python, Django) used by physicians. (130k+ records, 50k+ active users)  
- Led Data Science, AI, and Growth teams, helped securing 30M THB funding (Shark Tank Thailand S3).

**Facebook, Notification Backend** — Intern 2017  
- Implemented a pipeline for network effect prediction (classification/regression) using HiveQL and FB Learner Flow.  
- Increased comment metrics by 0.5% in production (2B active users).

**MIT Media Lab, Human Dynamics** — Undergrad Researcher, under Yan Leng 2018-2020  
- Developed an optimization framework (Python, Pandas) for learning network structure and marginal benefits from large-scale datasets (e.g., 400GB+ CDRs).

## publications

**LUSD: Localized Update Score Distillation for Text-Guided Image Editing** (co-first, *ICCV 2025*)  
W. Chinchuthakun\*, **T. Saengja**\*, N. Tritrong, P. Rewatbowornwong, P. Khungurn, S. Suwajanakorn. [arXiv:2503.11054](https://arxiv.org/abs/2503.11054)

**ZeroGuideSeg++: Improving Local Attention Masking in CLIP for Zero-Guidance Segmentation**  
P. Rewatbowornwong, N. Chatthee, **T. Saengja**, E. Chuangsuwanich, S. Suwajanakorn. (under review)

**Decomposing Food Images for Better Nutrition Analysis**  
P. Khlaisamniang, et al. (**T. Saengja** as corresponding author). *CVPR MetaFood 2025*

**GPU Poor's Guide to Radiology Report Generation**  
K. Udomlaksakul, et al. (**T. Saengja** as third author). *Proc. BioNLP 2024*.

## selected projects

**Accurate Clinical Decision Support Systems** (MIT 6.S897: ML for Healthcare, 2017)  
Analyzed drivers of clinical decision support dismissals using logistic regression, random forests, LDA.

**Paraphrase Identification** (MIT 6.867: Machine Learning, 2016)  
Designed BoW, word-embedding, SVM, NN, and LSTM models achieving 81.8% F1-score on MSR Paraphrase Corpus.